Parameter-Efficient Transfer Learning for Remote Sensing Image–Text Retrieval

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***Abstract*— Vision-and-language pretraining (VLP) models have experienced a surge in popularity recently. By fine-tuning them on specific datasets, significant performance improvements have been observed in various tasks. However, full fine-tuning of VLP models not only consumes a significant amount of computational resources but also has a significant environmental impact. Moreover, as remote sensing (RS) data are constantly being updated, full fine-tuning may not be practical for real-world applications. To address this issue, in this work, we investigate the parameter-efficient transfer learning (PETL) method to effectively and efficiently transfer visual–language knowledge from the natural domain to the RS domain on the image–text retrieval task. To this end, we make the following contributions. First, we construct a novel and sophisticated PETL framework for the RS image–text retrieval (RSITR) task, which includes the pretrained CLIP model, a multimodal RS adapter, and a hybrid multimodal contrastive (HMMC) learning objective. Second, to deal with the problem of high intramodal similarity in RS data, we design a simple yet effective HMMC loss. Third, we provide comprehensive empirical studies for PETL-based RSITR (PE-RSITR). Our results demonstrate that the proposed method is promising and of great potential for practical applications. Fourth, we benchmark extensive state-of-the-art (SOTA) PETL methods on the RSITR task. Our proposed model only contains 0.16 M training parameters, which can achieve a parameter reduction of 98.9% compared with full fine-tuning, resulting in substantial savings in training costs. Our retrieval performance exceeds traditional methods by 7%–13% and achieves comparable or better performance than full fine-tuning. This work can provide new ideas and useful insights for RS vision–language (VL) tasks.**

***Index Terms*— Adapter, cross-modal, parameter-efficient transfer learning (PETL), remote sensing image–text retrieval (RSITR).**

I. Introduction

With the development of Earth observation technology [1], remote sensing (RS) imagery is becoming more and more accessible, improving human's perception of the Earth [2], [3]. However, how to efficiently convert RS imagery into actionable information is still a significant research [4], [5], [6], [7]. To fully exploit the potential of RS images in human–computer interaction, RS vision–language (VL) tasks have become a hot research topic in recent years. The different granularity of VL multimodal tasks has been introduced into RS data, including image level [8], [9], [10], [11], object level [12], [13], [14], pixel level [15], and spatial–temporal level [16], [17]. These technologies have promising applications in urban planning, disaster monitoring, search and rescue activities, resource detection, and agricultural production [18], [19], [20], [21], [22], [23].

Large VL pretraining (VLP) models have surged [24], [25] in recent years. In particular, contrastive language-image pretraining (CLIP) [26] has shown great potential in multimodal representations and can project natural image and text modalities into a joint semantic subspace. As shown in Fig. 1, the aligned image–text samples have the same vector direction on the unit hypersphere. Fine-tuning large VLP models has become a fundamental paradigm of research.

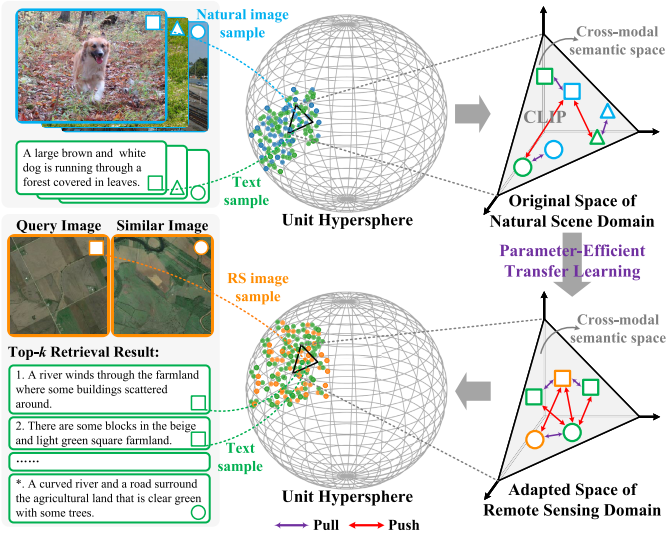


Fig. 1. Matched natural image–text samples have the same vector direction in the unit hypersphere space of the pretrained CLIP model. The PETL method learns specific knowledge of RS domain to get the adapted space. In the original space of the natural scene domain, only the distance between different modalities is paid attention to. However, in the RS domain, it is necessary to consider samples with high intramodal similarity to avoid the problem of matching errors.

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However, the research of transferring the knowledge learned from image–text pairs in the natural domain to a more complex RS domain is still underexplored.

Meanwhile, VLP models in the RS domain have not been proposed due to the challenges of RS VL tasks. Although there are a large number of publicly available RS images, few of them are captioned and even fewer are multicaptioned. The fully fine-tuned CLIP model has achieved encouraging performance for RS image classification.[1] However, this approach is not feasible due to the heavy computation, memory storage, and excessive CO_2 emissions. As RS images are updated, it is impractical to adapt with constant full fine-tuning on a daily basis. In this context, we explore a new research paradigm of parameter-efficient transfer learning (PETL) from the natural scene domain to the RS domain. Due to the outstanding representational capability of CLIP in VL, we use the CLIP model to study RS image–text retrieval (RSITR). This leads to a new research task, namely, PETL-based RSITR (PE-RSITR). The RSITR task can verify the performance of the adapted space of the RS domain, as shown in Fig. 1.

Nowadays, the mainstream methods of PETL are mainly divided into adapter [27] and prompt learning [28], [29]. PETL only fine-tunes a small number of parameters while keeping the parameters of the CLIP unchanged, which greatly reduces the computational cost while having comparable performance to full fine-tuning. However, the existing works typically focus on downstream tasks from the same domain of the VLP models. This creates a limit that a strong VLP model with sufficient knowledge may not be available in an unknown specific domain (e.g., RS). Therefore, there are still many challenges to exploring PE-RSITR.

First, CLIP is pretrained in the natural scene domain, which has domain gaps with the RS domain. To bridge the significant domain gap, the knowledge of CLIP needs to be transferred from "VL of the natural scene" to "VL of the RS." John von Neumann once said: with four parameters I can fit an elephant, and with five I can make him wiggle his trunk. Therefore, we attempt to design a method with a small number of trained parameters to explore the new knowledge of RS image–text efficiently while inheriting the prior knowledge structure of the natural scene domain appropriately. In addition, the RSITR task involves two modalities. If there is no cross-modal interaction mechanism, only suboptimal results can be obtained [30], [31]. Therefore, we further try to design a method that does not increase parameters and can accomplish cross-modal knowledge-sharing. Finally, the RS image–text data are very different from the data of the natural domain. The RS images are collected by satellites from an overhead view and the intraclass similarity is extremely high due to the Earth's texture. The visualization results of the textual similarity in the literature [10] show that the caption similarity is also high. Since this is not fully considered by the existing methods, RSITR often results in the error of misalignment of similar RS images or captions.

To tackle the above problems, we propose a novel and sophisticated PE-RSITR framework. Although adapter-based methods have been widely explored in prior works, it is still nontrivial to design an effective adapter for the RSITR task. Based on extensive experiments and explorations, we make the following design decisions: 1) we design a more compact multimodal RS adapter (MRS-Adapter) that has no skip connection and connects only once in parallel with the transformer block and 2) inspired by the cross-modal adapter [30], MRS-Adapter uses a linear layer for weight-sharing. The shared linear layer enables the fine-grained information of RS image modality and text modality to interact, which can enhance the RS VL modality representation.

Furthermore, token-level data augmentation is designed to construct intramodal positive pairs for RS images and texts. The method of data augmentation is to adopt the simple random dropout. By devising a simple yet efficient loss function of hybrid multimodal contrastive (HMMC) constraints without increasing parameters, the distance between the query image (query text) and other similar images (similar texts) can be pushed to avoid matching errors. We have conducted extensive experiments on three commonly used datasets, i.e., RSICD [11], RSITMD [10], and UCM [32]. First, we benchmark many state-of-the-art (SOTA) methods on the PE-RSITR task, and the adapter largely outperforms the prompt learning method. Second, the trained parameter of our proposed method is 0.16 M, which can reduce 98.9% parameters of full fine-tuning and greatly save the training cost. Finally, our retrieval performance exceeds the traditional methods by 7%–13% and achieves comparable or better than full fine-tuning. Our PE-RSITR framework is both parameter-efficient and effective.

In general, our contributions can be summarized in the following aspects.

1) We propose a novel and sophisticated PETL framework for the RSITR task. Specifically, the proposed framework consists of the pretrained CLIP model, the MRS-Adapter, and an HMMC learning objective.
2) We design a simple yet effective loss function: the HMMC loss for PE-RSITR. The experimental results prove that the proposed HMMC loss is effective in further improving the performance on top of the proposed MRS-Adapter.
3) We provide comprehensive empirical studies for the PE-RSITR task. Our qualitative and quantitative results demonstrate that the proposed method is promising and of great potential for practical applications.
4) Extensive experiments show that our approach can achieve a parameter reduction of 98.9% without performance sacrifice compared with full fine-tuning. Our performance exceeds the traditional methods by 7%–13%. The comprehensive benchmark results are insightful for future research.

This article is organized as follows. We review the related work of RSITR, the VLP model, and PETL in Section II. In Section III, we present our proposed PE-RSITR framework. Evaluation methods and extensive experiment results are shown in Section IV. Finally, we conclude this work in Section V.

[1] https://github.com/arampacha/CLIP-rsicd

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II. Related Work

A. Remote Sensing Image–Text Retrieval

With the development of RS VL cross-modal technology, RSITR is becoming a major interest. RSITR can effectively verify the performance of VL modal representations. However, due to the complexity of RS image–text data, there have been limited related works. Some works [33], [34], [35], [36] use CNN (e.g., VGG, ResNet) to extract RS image features and RNN (e.g., LSTM, BiLSTM) to extract text features. Hoxha et al. [34] first encoded the RS image and converted it into a caption, and finally calculated the similarity with the real captions to complete the matching. Rahhal et al. [35] proposed an unsupervised image–text retrieval method for RS imagery. To reduce the occupancy and overhead of the retrieval algorithm, Yuan et al. [9] proposed a lightweight RS multiscale cross-modal retrieval model (LW-MCR) and designed distillation loss and semisupervised loss to enhance the retrieval performance. Yuan et al. [10] also proposed an asymmetric multimodal multisource image retrieval method that uses the multiscale self-attention module to extract salient features of RS images and uses the features to guide the text representation. In the RSITR framework based on global and local information (GaLR), Yuan et al. [10] indicated that RSITR should focus not only on the global features of RS images but also on the local features reflecting object relationships and saliency. Recently, multilanguage transformer [37] demonstrated that loading CLIP pretraining model [26] can achieve promising performance on RSITR.

B. Vision–Language Pretraining Model

Large-scale VLP models are developing rapidly and have shown encouraging results on various downstream tasks [24]. According to the encoder type, VLP models are mainly classified into fusion encoder and dual encoder [25]. The fusion encoder takes image and text features as input and uses some fusion methods for VL interaction. The fusion encoder is mainly classified into single-stream and dual-stream structures. The single-stream structure (e.g., OSCAR [38], XGPT [39], SimVLM [40]) concatenates multimodal features and uses the transformer encoder in a unified framework. However, the single-stream performs the self-attention directly on two modalities, ignoring the intermodal interaction. Therefore, the dual-stream structure (e.g., ViLBERT [41], ALBEF [42]) performs the cross-attention using the transformer decoder. The fusion encoder relies on a large transformer for VL interaction modeling but the inference process can be very slow in solving matching tasks such as image–text retrieval. In contrast, the dual encoder (e.g., CLIP [26], ALIGN [43]) uses some simple methods for VL interaction modeling and calculates similarity scores after projecting the image and text features into the same semantic subspace. This method is more efficient for retrieval tasks, e.g., CLIP shows amazing results for the image–text retrieval, but does not work well when dealing with VL understanding tasks. The current pretraining models for RS image research have also been greatly developed [44]. Zhang et al. [45] propose a transfer learning method from natural scenes to the RS domain, which can achieve good results in many tasks (e.g., scene classification, object detection, and land cover classification). Unfortunately, there is no sufficiently large and uniform RS image–text dataset to support VLP models for the RS domain.

C. Parameter-Efficient Transfer Learning

The existing methods of PETL are broadly divided into two families, i.e., prompt learning and adapter, and are summarized in detail as follows.

1) Prompt Learning: Prompt learning [28], [29] was first proposed in natural language processing (NLP). When fine-tuning large language models, task-specific learnable vectors are added to the input. Unlike full fine-tuning, prompt learning can significantly reduce storage and computational costs and can achieve comparable performance. Prompt learning has been applied to computer vision (CV). Visual prompt tuning [46] (VPT) can efficiently fine-tune large-scale transformer models in vision. For CLIP-based image classification, Zhou et al. [47] added continuous learnable prompts on text labels for context optimization (CoOp) and then proposed to generate prompts using image features for conditional context optimization (CoCoOp) [48]. Zang et al. [49] further combined the advantages of text prompt learning and visual prompt learning to propose a tiny network to jointly optimize prompt learning for different modalities. MaPLe [50] used the V-L coupling function to generate visual prompts on textual prompts to adapt both language and vision branches simultaneously. However, there are few prompt learning methods for multimodal tasks. CPT [51] proposed a new prompt learning paradigm for visual grounding by adding color prompts to text and images, respectively. CPT used the color-masked token of the target region and color text prompts to ground the object.

2) Adapter: Since Houlsby et al. [27] proposed the adapter module to fine-tune large pretrained models in NLP, many improved methods [52], [53], [54], [55] have shown good performance on NLP tasks. Adapter is a lightweight plug-and-play module. The pretrained model is frozen during fine-tuning, and the parameters of the adapter are updated. Recently, the adapter is widely used to fine-tune pretrained models in CV. Chen et al. [56] proposed Conv-Adapter which replaces the original linear layers with convolutional layers, making it possible to efficiently fine-tune large-scale ConvNets. Convpass [57] is also composed of convolutional layers, but it is the adapter for vision transformer. AdapterFormer [58] is the adapter based on the original linear layers for vision transformer, and it can adapt both image and video tasks efficiently. Pan et al. [59] proposed a new spatio-temporal adapter (ST-Adapter) to accomplish PETL from image models to video tasks. With the appearance of CLIP, the adapter combined with CLIP further shows superior performance. CLIP-Adapter [60] uses an additional bottleneck layer with residual connections at the end of the image and text branches, respectively. Zhang et al. [61] proposed a training-free Tip-Adapter for CLIP for few-shot classification. SVL-Adapter [62] combines the complementary advantages of CLIP and self-supervised representation learning for image classification that are significantly different from common images. Previous works had

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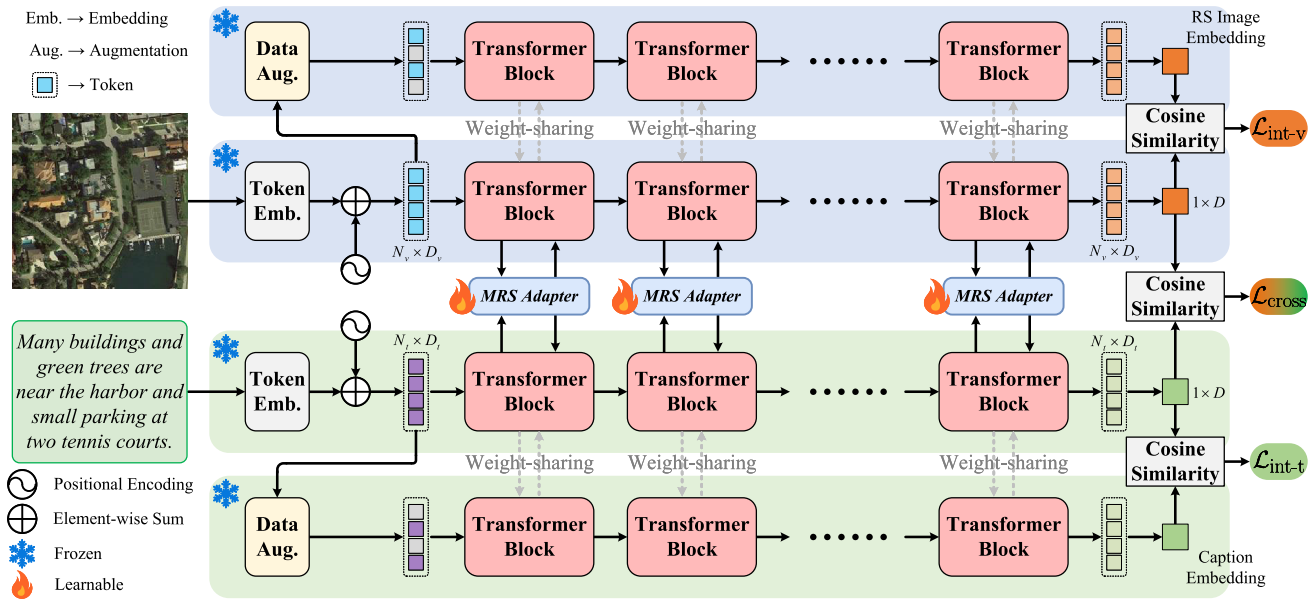


Fig. 2. Overall architecture of our proposed novel and sophisticated PE-RSITR framework. It mainly consists of three parts: the frozen CLIP pretraining backbone, the MRS-Adapter with cross-modal interaction, and an HMMC loss.

focused on unimodal tasks and less on multimodal tasks. Although VL-Adapter [63] is proposed for image–text and video–text multimodal tasks, it uses CLIP for image encoding and uses Adapter only in the language model. The latest cross-modal adapter [30] and UniAdapter [31] both propose cross-modal interaction mechanism and have shown good performance in multimodal tasks such as image/video text retrieval and visual question answering.

III. Methodology

To capitalize a large VLP model of natural scenes for RS VL tasks with domain differences, such as RSITR, the intrinsic gap in different domains needs to be filled. This section illustrates our proposed PE-RSITR framework in detail, and the overall framework is shown in Fig. 2.

A. Preliminary

In this section, we briefly describe how to process RS images and caption embeddings through the CLIP pretraining model and introduce the basic structure of the adapter.

1) Multimodal Encoder: We use the CLIP pretraining model as the primary multimodal encoder, including two branches of image encoder (ViT-B/32) and text encoder with the same structure. Concretely, given a cross-modal RS image–query dataset $\mathcal{O} = \{(i_n, t_n)\}_{n=1}^{N}$, where there are N pairs of image–text positive pairs (i_n, t_n). To simplify the notations, we denote $\boldsymbol{I} \in \mathbb{R}^{H\times W\times 3}$ and $T = \{w_m\}_{m=1}^{M}$ (M is the sentence length) as single instances of RS image and query text modality, respectively, where $H \times W \times 3$ denotes the size of the RS image and w_m represents the mth word. The basic architecture of the image encoder is shown in the blue background branch in Fig. 2. First, a convolution layer is used to generate the patch tokens $\boldsymbol{I}_{\text{patch}} \in \mathbb{R}^{(H/s)\cdot(W/s)\times D_v}$, where s is the stride of the backbone network and also the patch size. An additional classification token (CLS) token $\boldsymbol{I}_{\text{cls}} \in \mathbb{R}^{1\times D_v}$ is added to the token sequence, and the vision transformer adds the positional embedding $\boldsymbol{I}_{\text{pos}} \in \mathbb{R}^{N_v\times D_v}$ into each token. The RS image tokens are defined as

$$\boldsymbol{I}_0 = \left[\boldsymbol{I}_{\text{cls}}; \boldsymbol{I}_{\text{patch}}\right] + \boldsymbol{I}_{\text{pos}} \tag{1}$$

where $\boldsymbol{I}_0 \in \mathbb{R}^{N_v\times D_v}$, $N_v = 1 + (H/s)\cdot(W/s)$, and D_v is the hidden dimension of the vision transformer. Afterward, the RS image tokens $\boldsymbol{I}_0$ pass through 12 layers of stacked transformer blocks. The lth transformer block can be represented as

$$\hat{\boldsymbol{I}}_l = \text{MHA}(\text{LN}(\boldsymbol{I}_{l-1})) + \boldsymbol{I}_{l-1} \tag{2}$$

$$\boldsymbol{I}_l = \text{MLP}\big(\text{LN}\big(\hat{\boldsymbol{I}}_l\big)\big) + \hat{\boldsymbol{I}}_l \tag{3}$$

where $\hat{\boldsymbol{I}}_l$ and $\boldsymbol{I}_l$, respectively, indicate the output of multihead attention (MHA) and feedforward network (FNN) modules. Finally, the CLS token is used as the global RS image embedding $\boldsymbol{v}$ which is linearly projected into the D-dimensional cross-modal semantic space and then L_2 normalized.

The green background branch in Fig. 2 is the text encoder. The text encoder performs token embedding similarly. First, the caption is tokenized using lower-cased byte pair encoding (BPE), denoted as $\boldsymbol{T}_{\text{token}} \in \mathbb{R}^{N_t\times D_t}$. The token sequence for each caption starts with a [BOS] token and ends with a [EOS] token. Afterward, the positional embedding $\boldsymbol{T}_{\text{pos}} \in \mathbb{R}^{N_t\times D_t}$ is added to each token. The text tokens are defined as

$$\boldsymbol{T}_0 = \boldsymbol{T}_{\text{token}} + \boldsymbol{T}_{\text{pos}} \tag{4}$$

where $\boldsymbol{T}_0 \in \mathbb{R}^{N_t\times D_t}$, and D_t is the hidden dimension of the text transformer. After word embedding, the tokens $\boldsymbol{T}_0$ are sent to 12 layers of stacked transformer blocks. The lth transformer block can be represented as

$$\hat{\boldsymbol{T}}_l = \text{MHA}(\text{LN}(\boldsymbol{T}_{l-1})) + \boldsymbol{T}_{l-1} \tag{5}$$

$$\boldsymbol{T}_l = \text{MLP}\big(\text{LN}\big(\hat{\boldsymbol{T}}_l\big)\big) + \hat{\boldsymbol{T}}_l \tag{6}$$

where $\hat{\boldsymbol{T}}_l$ and $\boldsymbol{T}_l$, respectively, indicate the output of MHA and FNN modules. Finally, the highest layer of the transformer

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at the [EOS] token is used as the global caption embedding $\boldsymbol{t}$ which is linearly projected into the D-dimensional cross-modal semantic space and then L_2-normalized. The only difference between the image encoder and the text encoder is that the hidden dimension $D_v = 768$ while $D_t = 512$.

2) Adapter: Inspired by the success of Adapter [27] in NLP, more and more adapter-based methods have shown promising performance in both the CV and VL tasks. The adapter consists of a bottleneck that contains few parameters relative to the original pretraining model. Specifically, the adapter first uses a down-projection linear layer with parameters $\boldsymbol{W}_{\text{down}} \in \mathbb{R}^{d\times\hat{d}}$ ($\hat{d} \ll d$) to project the input features onto a low-dimensional representation. Then a nonlinear activation function is used, commonly the ReLU activation function. Finally, an up-projection linear layer with parameters $\boldsymbol{W}_{\text{up}} \in \mathbb{R}^{\hat{d}\times d}$ projects the features back to the input size before adding the skip connection. Formally, given an input feature $\boldsymbol{X} \in \mathbb{R}^{N_{\text{in}}\times d}$, the adapted feature $\tilde{\boldsymbol{X}} \in \mathbb{R}^{N_{\text{in}}\times d}$ can be calculated as

$$\boldsymbol{X}^{\text{down}} = \text{ReLU}(\boldsymbol{X}\cdot\boldsymbol{W}_{\text{down}}) \tag{7}$$

$$\text{Adapter}(\boldsymbol{X}) = \tilde{\boldsymbol{X}} = s\cdot\boldsymbol{X}^{\text{down}}\cdot\boldsymbol{W}_{\text{up}} + \boldsymbol{X} \tag{8}$$

where s is a scalar scale factor that controls the effect of the adapter. The original adapter scheme is inserted sequentially into the MHA and FFN of the transformer block. The feature adaptation process at the lth layer can be written as

$$\hat{\boldsymbol{X}}_l = \text{Adapter}(\text{MHA}(\text{LN}(\boldsymbol{X}))) + \boldsymbol{X} \tag{9}$$

$$\boldsymbol{X}_l = \text{Adapter}\big(\text{MLP}\big(\text{LN}\big(\hat{\boldsymbol{X}}_l\big)\big)\big) + \hat{\boldsymbol{X}}_l. \tag{10}$$

B. MRS-Adapter

Various CLIP-based adapter methods show great potential for VL tasks. The core of extending CLIP to the RS domain to accomplish RSITR lies in exploring the VL expert knowledge in the RS domain efficiently while appropriately inheriting the VL prior knowledge structure of the natural scene domain. This work is still underexplored. If the adapter is directly extended from NLP to both the modalities of VL, it can only lead to suboptimal results due to the lack of cross-modal interactions. This point was also verified in the recent works [30], [31]. Therefore, we attempt to design an adapter that can share knowledge between RS image modality and text modality without increasing parameters. Finally, we found that the same cross-modal interaction as the cross-modal adapter is the most effective way and can reduce parameters. Our MRS-Adapter is extremely similar to the cross-modal adapter, but our scheme is more concise. MRS-Adapter has no skip connection and is only connected in parallel with the FFN module, which can further reduce the number of adapters. The specific structure is shown in Fig. 3. Formally, the input of the lth layer is the $\hat{\boldsymbol{I}}_l \in \mathbb{R}^{N_v\times D_v}$ and $\hat{\boldsymbol{T}}_l \in \mathbb{R}^{N_t\times D_t}$ of the MHA module output, and the adapted features can be obtained as follows:

$$\hat{\boldsymbol{I}}_l^{\text{down}} = \text{ReLU}\big(\hat{\boldsymbol{I}}_l\cdot\boldsymbol{W}_{\text{down}}^{v}\big) \tag{11}$$

$$\hat{\boldsymbol{I}}_l^{\text{MRS-Adapter}} = \Big[\hat{\boldsymbol{I}}_l^{\text{down}}\cdot\boldsymbol{W}_{\text{up}}^{v};\quad \hat{\boldsymbol{I}}_l^{\text{down}}\cdot\boldsymbol{W}_{\text{up}}^{\text{share}}\Big] \tag{12}$$

$$\hat{\boldsymbol{T}}_l^{\text{down}} = \text{ReLU}\big(\hat{\boldsymbol{T}}_l\cdot\boldsymbol{W}_{\text{down}}^{t}\big) \tag{13}$$

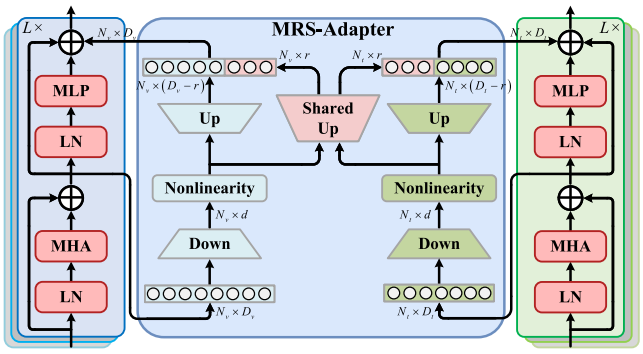


Fig. 3. Implementation details of our MRS-Adapter. MRS-Adapter is inserted in parallel with the FFN module and adds a modality-shared up-projection to connect the original modality-specific up-projection linear layer for cross–modal weight-sharing.

$$\hat{\boldsymbol{T}}_l^{\text{MRS-Adapter}} = \Big[\hat{\boldsymbol{T}}_l^{\text{down}}\cdot\boldsymbol{W}_{\text{up}}^{t};\quad \hat{\boldsymbol{T}}_l^{\text{down}}\cdot\boldsymbol{W}_{\text{up}}^{\text{share}}\Big] \tag{14}$$

where $d \ll D_v$, $d \ll D_t$, $0 < r < D_v$, $0 < r < D_t$, $\boldsymbol{W}_{\text{down}}^{v} \in \mathbb{R}^{D_v\times d}$ and $\boldsymbol{W}_{\text{down}}^{t} \in \mathbb{R}^{D_t\times d}$ are the modality-specific down-projection weights of two branches, $\boldsymbol{W}_{\text{up}}^{\text{share}} \in \mathbb{R}^{d\times r}$ are the modality-shared weights, and $\boldsymbol{W}_{\text{up}}^{v} \in \mathbb{R}^{d\times D_v - r}$ and $\boldsymbol{W}_{\text{up}}^{t} \in \mathbb{R}^{d\times D_t - r}$ are the modality-specific up-projection weights of two branches. Finally, the MRS-Adapter is connected in parallel with the FFN module at each layer of both the branches, and the feature adaptation process at the lth layer can be written as

$$\hat{\boldsymbol{I}}_l = \text{MHA}(\text{LN}(\boldsymbol{I}_{l-1})) + \boldsymbol{I}_{l-1} \tag{15}$$

$$\boldsymbol{I}_l = \text{MLP}\big(\text{LN}\big(\hat{\boldsymbol{I}}_l\big)\big) + \hat{\boldsymbol{I}}_l + \hat{\boldsymbol{I}}_l^{\text{MRS-Adapter}} \tag{16}$$

$$\hat{\boldsymbol{T}}_l = \text{MHA}(\text{LN}(\boldsymbol{T}_{l-1})) + \boldsymbol{T}_{l-1} \tag{17}$$

$$\boldsymbol{T}_l = \text{MLP}\big(\text{LN}\big(\hat{\boldsymbol{T}}_l\big)\big) + \hat{\boldsymbol{T}}_l + \hat{\boldsymbol{T}}_l^{\text{MRS-Adapter}}. \tag{18}$$

MRS-Adapter adding an r-dimensional linear layer for weight-sharing can directly reduce $d \times r$ parameters. The shared up-projection enables the fine-grained information of RS image modality and text modality to interact, which can enhance the RS VL modality representation. MRS-Adapter can learn the VL knowledge specific to the RS domain and efficiently extend the natural scene domain to the RS domain.

C. HMMC Learning Objective

In the image–text retrieval tasks, the bidirectional triplet loss established by Faghri et al. [64] has become the mainstream loss function. The bidirectional triplet loss can pull the distance between this sample and the positive sample of another modality closer while pushing the distance between it and the negative sample of another modality farther. However, this framework can only constrain the intermodal samples and does not consider the intramodal samples. In particular, RS images are characterized by extremely high intraclass similarity, and the visualization results of the textual similarity in the literature [10] show that the textual similarity is also extremely high. Therefore, RSITR often results in the error of retrieving misalignment of similar RS images or captions. To cope with the problem, token-level data augmentation is adopted to construct intramodal positive pairs for RS images

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and texts. We design a simple yet effective HMMC loss function. The framework of intermodal and intramodal cooperative constraints is shown in Fig. 2. Inspired by contrastive learning [65], the method of data augmentation is to adopt the simple random dropout.

1) Intramodal Constraints: The bidirectional triplet loss with hard negatives is used, and the similarity between sample pairs is measured by cosine similarity. For RS image modality, data augmentation is performed after token embedding, and a token-level positive pair $(\boldsymbol{I}_0, \boldsymbol{I}_0^+)$ is obtained for each RS image $\boldsymbol{I}$. The process is denoted as

$$\boldsymbol{I}_0^+ = \text{random_dropout}(\boldsymbol{I}_0). \tag{19}$$

Then we perform the intramodal constraint of the RS image to pull the distance from the query image and other similar images. We denote the final embedding of $\boldsymbol{I}_0^+$ as $\boldsymbol{v}^+$. For a positive pair $(\boldsymbol{v}, \boldsymbol{v}^+)$, the intramodal triplet loss we adopt is

$$\begin{aligned}\mathcal{L}_{\text{intra-v}}(\boldsymbol{v}, \boldsymbol{v}^+) = &\sum_{\widehat{\boldsymbol{v}}^+}\left[\alpha_v - \cos(\boldsymbol{v}, \boldsymbol{v}^+) + \cos(\boldsymbol{v}, \widehat{\boldsymbol{v}}^+)\right]_+ \\ &+ \sum_{\widehat{\boldsymbol{v}}}\left[\alpha_v - \cos(\boldsymbol{v}, \boldsymbol{v}^+) + \cos(\widehat{\boldsymbol{v}}, \boldsymbol{v}^+)\right]_+\end{aligned} \tag{20}$$

where α_v is the margin of the intramodal constraint of the RS image and $[x]_+ = \max(x, 0)$. Similarly, the text branch also uses random dropout to generate a token-level positive pair $(\boldsymbol{T}_0, \boldsymbol{T}_0^+)$ for each caption $\boldsymbol{T}$, calculated as follows:

$$\boldsymbol{T}_0^+ = \text{random_dropout}(\boldsymbol{T}_0). \tag{21}$$

Then we perform the intramodal constraint of the caption text to pull the distance from the query caption and other similar captions. We denote the final embedding of $\boldsymbol{T}_0^+$ as $\boldsymbol{t}^+$. For a positive pair $(\boldsymbol{t}, \boldsymbol{t}^+)$, the intramodal triplet loss is

$$\begin{aligned}\mathcal{L}_{\text{intra-t}}(\boldsymbol{t}, \boldsymbol{t}^+) = &\sum_{\widehat{\boldsymbol{t}}^+}\left[\alpha_t - \cos(\boldsymbol{t}, \boldsymbol{t}^+) + \cos\left(\boldsymbol{t}, \widehat{\boldsymbol{t}}^+\right)\right]_+ \\ &+ \sum_{\widehat{\boldsymbol{t}}}\left[\alpha_t - \cos(\boldsymbol{t}, \boldsymbol{t}^+) + \cos(\widehat{\boldsymbol{t}}, \boldsymbol{t}^+)\right]_+\end{aligned} \tag{22}$$

where α_t is the margin of the intramodal constraint of the caption text.

2) Cross-Modal Constraint: The multimodal alignment of RS image–text is promoted by relying on the global similarity of RS image–text. We compute the cross-modal constraint loss with

$$\begin{aligned}\mathcal{L}_{\text{cross}}(\boldsymbol{v}, \boldsymbol{t}) = &\sum_{\widehat{\boldsymbol{t}}}\left[\lambda - \cos(\boldsymbol{v}, \boldsymbol{t}) + \cos(\boldsymbol{v}, \widehat{\boldsymbol{t}})\right]_+ \\ &+ \sum_{\widehat{\boldsymbol{v}}}[\lambda - \cos(\boldsymbol{v}, \boldsymbol{t}) + \cos(\widehat{\boldsymbol{v}}, \boldsymbol{t})]_+\end{aligned} \tag{23}$$

where λ is the margin of the cross-modal constraint.

3) Overall Objective: By combining hard-negative-based intramodal constraints loss with cross-modal constraints loss, we obtain the HMMC loss

$$\mathcal{L}(\boldsymbol{v}, \boldsymbol{t}) = \mathcal{L}_{\text{cross}}(\boldsymbol{v}, \boldsymbol{t}) + \mathcal{L}_{\text{intra-v}}(\boldsymbol{v}, \boldsymbol{v}^+) + \mathcal{L}_{\text{intra-t}}(\boldsymbol{t}.\boldsymbol{t}^+). \tag{24}$$

IV. Experiments

In this section, we first describe the dataset, evaluation metrics, and experimental setup details in Sections IV-A and IV-B. Further, Section IV-C introduces the SOTA approaches and provides comparisons of retrieval performance. Section IV-D conducts result analyses. In Section IV-E, we perform sufficient ablation experiments. Finally, we present some visualization results to further analyze in Section IV-F.

A. Dataset and Evaluation Metrics

We evaluate our proposed PE-RSITR framework on the three widely used RS image–text datasets: RSICD [11], RSITMD [10], and UCM [32]. RSICD is the dataset with the largest number of samples, while RSITMD is the dataset with more fine-grained captions and more challenges. UCM requires the model to be robust because of the small numbers.

Two evaluation metrics Recall at K ($R@K$, $K = 1$, 5, and 10) and mean recall (mR) are exploited to assess our model. $R@K$ aims to calculate the ratio of queries that successfully retrieve the ground truth as one of the first K results. mR represents the average of $R@K$ for both the text retrieval and image retrieval, which evaluates the overall retrieval performance and can be formulated in the equation below

$$\text{mR} = \left(\underbrace{R@1+R@5+R@10}_{\text{Text retrieval}} + \underbrace{R@1+R@5+R@10}_{\text{Image retrieval}}\right)/6. \tag{25}$$

B. Implementation Details

All the experiments in this work are conducted on one NVIDIA RTX 3090 24-GB GPU. We follow the data partitioning approach of Yuan et al. [10] and use 80%, 10%, and 10% of the dataset as the training set, validation set, and test set, respectively. For the RS image, we resize the image size to a fixed size of 224 × 224 for training. We set the dimension d and r to 64 and the probability of random dropout to 0.2. The margin λ, α_v, and α_t are set to 0.2 for the triplet loss calculation. We set the initial learning rate of our network to 0.0002 for trained parameters and weight decay by 0.7 every 20 epochs. During training, we adopt the Adam optimizer to train our network with a batch size of 16 for 30 epochs. To make the experiment more convincing, we follow the works in GaLR [8] and MCRN [66] to conduct the experiments and report results. We leverage k-fold cross-validation to obtain an average result, and k is set to 5.

C. Comparisons With State-of-the-Art Methods

In this experiment, we comprehensively compare our proposed method with the traditional cross-modal retrieval methods and CLIP-based methods.

1) Traditional Methods: Following the previous literature, we also compare the proposed method with the progressive image–text retrieval models (VSE++ [64], SCAN [67], CAMP [68], MTFN [69], LW-MCR [9], AMFMN [10], GaLR [8]) on three RS image–text datasets. For these methods,

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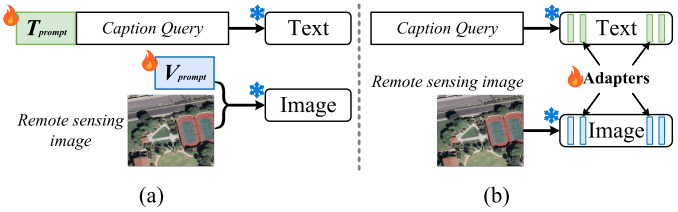


Fig. 4. (a) Prepending a sequence of learnable prompt tokens to the input tokens of the visual or text encoder. Only these added prompt tokens are updated during fine-tuning. (b) Insert a lightweight adapter into the visual or text encoder and update the parameters of the adapter during fine-tuning.

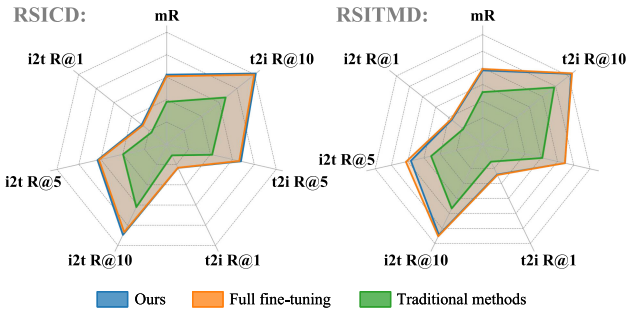


Fig. 5. Comparison to the RSITR results of full fine-tuning and the best results of traditional methods on the RSICD and RSITMD datasets.

we use the results in three literature [8], [9], [10]. In addition, we have added two latest RSITR methods.

1) *MCRN [66]:* MCRN constructs a multisource cross-modal retrieval network capable of image modality, text modality, and audio modality alignment based on shared networks of pattern memory and generative adversarial theory.
2) *CABIR [70]:* CABIR proposes a cross-attention model based on region-level semantic features of RS images, with textual semantics to allocate weights and filter redundant features for image regions.

2) CLIP-Based Methods: Single Language [37] is loaded with CLIP pretraining parameters for training. In addition, we benchmark extensive efficient and commonly used PETL approaches in our PE-RSITR task. To assess the merits of our proposed method, we report our performance and compare it with the following methods.

1) *Zero-Shot CLIP [26]:* Testing directly on the sample of the unseen RS domain.
2) *Linear Probe:* Adding an extra linear layer on top of each of the two branches of the backbone and freezing all the parameters except the parameters in the linear layer.
3) *Full Fine-Tuning:* Fully updating all the parameters. In the experiments of the RSICD dataset, full fine-tuning uses the weight of the CLIP-rsicd model.[1]
4) *Prompt Learning:* As shown in Fig. 4(a), prepending a sequence of learnable prompt tokens to the input tokens, and only these added prompt tokens are updated during fine-tuning. Specifically, we compare CoOp [47] (added in the text tokens) and VPT [46] (added in the visual tokens). Following Jiang et al. [30], we applied both the visual and text tokens, called VL-Prompt.
5) *Adapter Tuning:* Adapter is a lightweight plug-and-play module. The pretrained model is frozen during fine-tuning, and the parameters of adapters are updated, as shown in Fig. 4(b). CLIP-Adapter [60] uses an additional bottleneck layer to learn new features and make residual connections. Adapter [27] is for the language transformer, Convpass [57] is the convolutional adapter for ViT, and AdaptFormer [58] is for adapting ViT to different image and video tasks. UniAdapter [31] and cross-modal adapter [30] are multimodal adapters. UniAdapter cannot be used directly due to the different dimensions of the visual branch and text branch. We use a linear layer to project visual features to 512 dimensions.

D. Results' Analysis

1) Results on Three Datasets: Tables I–III show the test results on the three datasets: RSICD, RSITMD, and UCM. We present results using various traditional methods and CLIP-based fine-tuning methods. Fig. 5 shows the retrieval results of full fine-tuning and the best results of the traditional methods on the RSICD and RSITMD datasets. Our method outperforms the traditional methods by 7%–13% and even surpasses the full fine-tuning. CLIP possesses a more complex and larger network structure with a higher number of parameters. Compared with traditional methods, PETL, based on CLIP, can achieve significant performance gains by leveraging the powerful generalization ability and rich visual–linguistic prior knowledge obtained through pretraining on massive datasets of natural scenes.

We provide comprehensive empirical studies for PE-RSITR. We observe that our PE-RSITR framework outperforms other works on the RSICD dataset. Compared with full fine-tuning, our results are improved by about 1%. Currently, the RSITMD dataset is the highest fine-grained and most challenging RS image–text data. Our approach can obtain optimal and suboptimal results, achieving comparable results to full fine-tuning. The UCM dataset challenges the robustness of the model due to its small size. As the results are shown in Table III, our method has the most optimal results and exceeds the effect of full fine-tuning. This result demonstrates the robustness of our PE-RSITR framework.

Single language adopts the dual-transformer structure and loads the pretrained parameters of CLIP to train. It achieves an average improvement of 8% in retrieval performance over the best traditional methods on the three datasets. Zero-shot CLIP, linear probe, and full fine-tuning serve as the fundamental baselines for PETL. Zero-shot CLIP directly uses the pretrained model for testing, resulting in the poorest performance on the three datasets. Full fine-tuning updates all the parameters and theoretically yields optimal results. However, due to the limited data size of UCM, full fine-tuning tends to overfit and fails to achieve the best performance, as shown in Table III.

2) Performance Analysis on Prompt Learning: In prompt learning, the VPT achieves superior performance, surpassing

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TABLE I

RETRIEVAL PERFORMANCE OF SOTA METHODS ON THE RSICD TEST SET. THE BEST PERFORMANCE OF TRADITIONAL METHODS IS WITH BOLD. RED AND BLUE REPRESENT THE BEST AND SECOND PERFORMANCE IN CLIP-BASED AND TRANSFER LEARNING METHODS, RESPECTIVELY

Methods	Reference	Backbone (image/text)	Params	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods										
VSE++ [64]	*BMVC'18*	ResNet18, GRU	15.78M	3.38	9.51	17.46	2.82	11.32	18.10	10.43
SCAN t-i [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	4.39	10.90	17.64	3.91	16.20	26.49	13.25
SCAN i-t [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	5.85	12.89	19.84	3.71	16.40	26.73	14.23
CAMP-triplet [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	5.12	12.89	21.12	4.15	15.23	27.81	14.39
CAMP-bce [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	4.20	10.24	15.45	2.72	12.76	22.89	11.38
MTFN [69]	*ACM MM'19*	Faster R-CNN, biGRU	77.90M	5.02	12.52	19.74	4.90	17.17	29.49	14.81
LW-MCR-b [9]	*TGRS'22*	SqueezeNet, -	1.65M	4.57	13.71	20.11	4.02	16.47	28.23	14.52
LW-MCR-d [9]	*TGRS'22*	SqueezeNet, -	1.65M	3.29	12.52	19.93	4.66	17.51	30.02	14.66
LW-MCR-u [9]	*TGRS'22*	SqueezeNet, -	1.65M	4.39	13.35	20.29	4.30	18.85	32.34	15.59
AMFMN-soft [10]	*TGRS'22*	ResNet18, biGRU	35.94M	5.05	14.53	21.57	5.05	19.74	31.04	16.02
AMFMN-fusion [10]	*TGRS'22*	ResNet18, biGRU	35.94M	5.39	15.08	23.40	4.90	18.28	31.44	16.42
AMFMN-sim [10]	*TGRS'22*	ResNet18, biGRU	35.94M	5.21	14.72	21.57	4.08	17.00	30.60	15.53
MCRN [66]	*JAG'22*	ResNet18, biGRU	52.35M	6.59	19.40	30.28	5.03	19.38	32.99	18.95
CABIR [70]	*AS'22*	ResNet152, BERT+biGRU	-	**8.59**	16.27	24.13	**5.42**	**20.77**	**33.58**	18.12
GaLR w/o MR [8]	*TGRS'22*	ResNet18, biGRU	46.89M	6.50	18.91	29.70	5.11	19.57	31.92	18.62
GaLR with MR [8]	*TGRS'22*	ResNet18, biGRU	46.89M	6.59	**19.85**	**31.04**	4.69	19.48	32.13	**18.96**
CLIP-based methods										
Single Language [37]	*JSTARS'22*	CLIP(ViT-B-32)	151M	10.70	29.64	41.53	9.14	28.96	44.59	27.42
Zero-shot CLIP [26]	*ICML'21*	CLIP(ViT-B-32)	0.00M	6.77	15.37	23.15	5.01	15.75	24.21	15.04
Linear probe [26]	*ICML'21*	CLIP(ViT-B-32)	1.05M	8.46	24.41	37.72	7.81	25.89	42.47	24.46
Full fine-tuning[1]	-	CLIP(ViT-B-32)	151M	**13.54**	**30.83**	**43.46**	**11.55**	**33.14**	**49.83**	**30.39**
CoOp [47]	*IJCV'22*	CLIP(ViT-B-32)	0.10M	6.32	15.89	27.60	4.31	17.89	31.73	17.29
VPT [46]	*ECCV'22*	CLIP(ViT-B-32)	0.46M	7.23	19.40	30.77	6.94	25.26	40.73	21.72
VL-Prompt	-	CLIP(ViT-B-32)	0.47M	6.47	17.05	28.68	6.69	22.62	36.16	19.61
Adapter [27]	*ICML'19*	CLIP(ViT-B-32)	0.17M	8.73	24.73	37.81	8.43	26.02	43.33	24.84
CLIP-Adapter [60]	*arXiv'21*	CLIP(ViT-B-32)	0.52M	7.11	19.48	31.01	7.67	24.87	39.73	21.65
Convpass [57]	*arXiv'22*	CLIP(ViT-B-32)	0.14M	6.54	19.67	32.78	7.03	23.08	39.15	21.38
AdaptFormer [58]	*NIPS'22*	CLIP(ViT-B-32)	0.17M	12.46	28.49	41.86	9.09	29.89	46.81	28.10
Cross-Modal Adapter [30]	*arXiv'22*	CLIP(ViT-B-32)	0.16M	11.18	27.31	40.62	9.57	30.74	48.36	27.96
UniAdapter [31]	*arXiv'23*	CLIP(ViT-B-32)	0.55M	12.65	30.81	42.74	9.61	30.06	47.16	28.84
Ours	-	CLIP(ViT-B-32)	0.16M	**14.13**	**31.51**	**44.78**	**11.63**	**33.92**	**50.73**	**31.12**

traditional methods. The CoOp and VL-Prompt can only achieve performance similar to traditional methods. Due to the significant visual feature gaps between natural and RS domains, adding learnable prompts into visual tokens (VPT) results in a 3%–5% higher retrieval performance compared with text tokens (CoOp). However, the VL-Prompt directly adds prompts to both the modalities, without considering multimodal interaction. The increase in prompts makes it more susceptible to overfitting and reduces its generalization ability.

Prompt length is the only additional hyperparameter needed to tune for prompt learning compared with full fine-tuning. We explore the impact of different prompt lengths on this task and analyze the performance of prompt learning in detail. Specifically, we follow the setting of CoOp [47] to vary the context length from 4 to 8 to 16 on the RSICD and RSITMD datasets. "end" or "mid" means putting the learnable token in the end or middle. The results are shown in Fig. 6, which shows that the learnable token in the middle position is better than the end, and more context tokens lead to better performance. However, too many tokens may lead to overfitting. Likewise, we follow the setting of VPT [46] to vary the prompt length from 1 to 100 to explore two variants: VPT-deep and VPT-shallow. As shown in Fig. 7, VPT-deep significantly outperforms VPT-shallow. The prompt length has a large impact on the VPT-deep results and shows that the optimal prompt length is 50 for different datasets. More prompts will cause overfitting and more resource consumption.

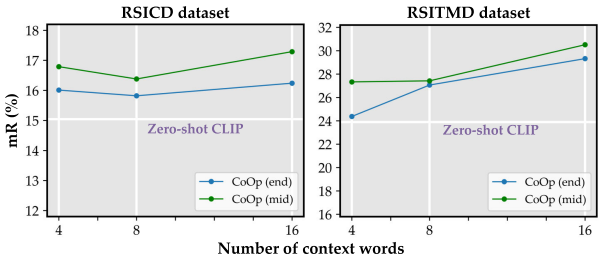


Fig. 6. Ablation on CoOp's context length. We vary the number of the context length for CoOp-end and CoOp-mid.

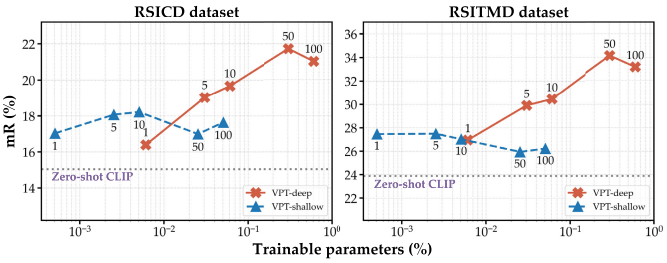


Fig. 7. Ablation on VPT's prompt length. We vary the number of prompts for VPT-deep and VPT-shallow.

3) Performance Analysis on Adapter: Observing the results in Tables I–III, we analyze the performance of adapter tuning. CLIP-Adapter [60] only adds a bottleneck layer at the end of the CLIP, making the transfer to the RS domain insufficient. The Adapter [27] for language models can only tune textual

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TABLE II

RETRIEVAL PERFORMANCE OF SOTA METHODS ON THE RSITMD TEST SET. THE BEST PERFORMANCE OF TRADITIONAL METHODS IS WITH BOLD. RED AND BLUE REPRESENT THE BEST AND SECOND PERFORMANCE IN CLIP-BASED AND TRANSFER LEARNING METHODS, RESPECTIVELY

Methods	Reference	Backbone (image/text)	Params	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods										
VSE++ [64]	*BMVC'18*	ResNet18, GRU	15.78M	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN t-i [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	10.18	28.53	38.49	10.10	28.98	43.53	26.64
SCAN i-t [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	11.06	25.88	39.38	9.82	29.38	42.12	26.28
CAMP-triplet [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	11.73	26.99	38.05	8.27	27.79	44.34	26.20
CAMP-bce [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN [69]	*ACM MM'19*	Faster R-CNN, biGRU	77.90M	10.40	27.65	36.28	9.96	31.37	45.84	26.92
LW-MCR-b [9]	*TGRS'22*	SqueezeNet, -	1.65M	9.07	22.79	38.05	6.11	27.74	49.56	25.55
LW-MCR-d [9]	*TGRS'22*	SqueezeNet, -	1.65M	10.18	28.98	39.82	7.79	30.18	49.78	27.79
LW-MCR-u [9]	*TGRS'22*	SqueezeNet, -	1.65M	9.73	26.77	37.61	9.25	34.07	54.03	28.58
AMFMN-soft [10]	*TGRS'22*	ResNet18, biGRU	35.94M	11.06	25.88	39.82	9.82	33.94	51.90	28.74
AMFMN-fusion [10]	*TGRS'22*	ResNet18, biGRU	35.94M	11.06	29.20	38.72	9.96	34.03	52.96	29.32
AMFMN-sim [10]	*TGRS'22*	ResNet18, biGRU	35.94M	10.63	24.78	41.81	**11.51**	34.69	**54.87**	29.72
MCRN [66]	*JAG'22*	ResNet18, biGRU	52.35M	13.27	29.42	41.59	9.42	35.53	52.74	30.33
GaLR w/o MR [8]	*TGRS'22*	ResNet18, biGRU	46.89M	13.05	30.09	**42.70**	10.47	36.34	53.35	31.00
GaLR with MR [8]	*TGRS'22*	ResNet18, biGRU	46.89M	**14.82**	**31.64**	42.48	11.15	**36.68**	51.68	**31.41**
CLIP-based methods										
Single Language [37]	*JSTARS'22*	CLIP(ViT-B-32)	151M	19.69	40.26	54.42	17.61	49.73	66.59	41.38
Zero-shot CLIP [26]	*ICML'21*	CLIP(ViT-B-32)	0.00M	9.29	26.33	37.39	7.79	23.67	38.89	23.89
Linear probe [26]	*ICML'21*	CLIP(ViT-B-32)	1.05M	17.02	33.12	48.35	13.33	41.80	63.89	36.25
Full fine-tuning	-	CLIP(ViT-B-32)	151M	**24.16**	**47.12**	**61.28**	**20.40**	**50.53**	**68.54**	**45.33**
CoOp [47]	*IJCV'22*	CLIP(ViT-B-32)	0.10M	12.19	30.69	42.82	9.16	33.85	54.35	30.51
VPT [46]	*ECCV'22*	CLIP(ViT-B-32)	0.46M	14.98	32.05	40.15	15.97	41.35	60.35	34.14
VL-Prompt	-	CLIP(ViT-B-32)	0.47M	12.81	31.28	42.64	12.61	36.20	58.84	32.40
Adapter [27]	*ICML'19*	CLIP(ViT-B-32)	0.17M	13.75	27.64	39.96	12.89	40.09	59.91	32.37
CLIP-Adapter [60]	*arXiv'21*	CLIP(ViT-B-32)	0.52M	12.83	28.84	39.05	13.30	40.20	60.06	32.38
Convpass [57]	*arXiv'22*	CLIP(ViT-B-32)	0.14M	16.03	30.16	40.26	12.05	38.66	58.11	32.55
AdaptFormer [58]	*NIPS'22*	CLIP(ViT-B-32)	0.17M	16.71	30.16	42.91	14.27	41.53	61.46	34.81
Cross-Modal Adapter [30]	*arXiv'22*	CLIP(ViT-B-32)	0.16M	18.16	36.08	48.72	16.31	44.33	64.75	38.06
UniAdapter [31]	*arXiv'23*	CLIP(ViT-B-32)	0.55M	19.86	36.32	51.28	17.54	44.89	65.46	39.23
Ours	-	CLIP(ViT-B-32)	0.16M	**23.67**	**44.07**	**60.36**	**20.10**	**50.63**	**67.97**	**44.47**

TABLE III

RETRIEVAL PERFORMANCE OF SOTA METHODS ON THE UCM TEST SET. THE BEST PERFORMANCE OF TRADITIONAL METHODS IS WITH BOLD. RED AND BLUE REPRESENT THE BEST AND SECOND PERFORMANCE IN CLIP-BASED AND TRANSFER LEARNING METHODS, RESPECTIVELY

Methods	Reference	Backbone (image/text)	Params	Text retrieval			Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional methods										
VSE++ [64]	*BMVC'18*	ResNet18, GRU	15.78M	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN t-i [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	14.29	45.71	67.62	12.76	50.38	77.24	44.67
SCAN i-t [67]	*ECCV'18*	Faster R-CNN, biGRU	13.68M	12.85	47.14	69.52	12.48	46.86	71.71	43.43
CAMP-triplet [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	10.95	44.29	65.71	9.90	46.19	76.29	42.22
CAMP-bce [68]	*ICCV'19*	Faster R-CNN, biGRU	27.03M	14.76	46.19	67.62	11.71	47.24	76.00	43.92
MTFN [69]	*ACM MM'19*	Faster R-CNN, biGRU	77.90M	10.47	47.62	64.29	**14.19**	52.38	78.95	44.65
LW-MCR-b [9]	*TGRS'22*	SqueezeNet, -	1.65M	12.38	43.81	59.52	12.00	46.38	72.48	41.10
LW-MCR-d [9]	*TGRS'22*	SqueezeNet, -	1.65M	15.24	**51.90**	62.86	11.90	50.95	75.24	44.68
LW-MCR-u [9]	*TGRS'22*	SqueezeNet, -	1.65M	**18.10**	47.14	63.81	13.14	50.38	79.52	45.35
AMFMN-soft [10]	*TGRS'22*	ResNet18, biGRU	35.94M	12.86	**51.90**	66.67	**14.19**	51.71	78.48	45.97
AMFMN-fusion [10]	*TGRS'22*	ResNet18, biGRU	35.94M	16.67	45.71	68.57	12.86	53.24	79.43	46.08
AMFMN-sim [10]	*TGRS'22*	ResNet18, biGRU	35.94M	14.76	49.52	68.10	13.43	51.81	76.48	45.68
CABIR [70]	*AS'22*	ResNet152, BERT+biGRU	-	15.17	45.71	**72.85**	12.67	**54.19**	**89.23**	**48.30**
CLIP-based methods										
Single Language [37]	*JSTARS'22*	CLIP(ViT-B-32)	151M	**19.04**	53.33	77.61	**19.33**	**64.00**	91.42	54.12
Zero-shot CLIP [26]	*ICML'21*	CLIP(ViT-B-32)	0.00M	10.95	34.76	55.71	7.33	35.14	54.57	33.08
Linear probe [26]	*ICML'21*	CLIP(ViT-B-32)	1.05M	13.33	52.71	77.62	14.43	59.42	90.28	51.30
Full fine-tuning	-	CLIP(ViT-B-32)	151M	17.14	**55.24**	79.52	13.90	56.95	91.81	52.43
CoOp [47]	*IJCV'22*	CLIP(ViT-B-32)	0.10M	7.19	42.76	74.19	9.78	48.34	87.41	44.95
VPT [46]	*ECCV'22*	CLIP(ViT-B-32)	0.46M	8.86	47.81	78.62	13.64	55.61	94.05	49.77
VL-Prompt	-	CLIP(ViT-B-32)	0.47M	7.97	45.48	75.96	12.51	51.57	91.52	47.50
Adapter [27]	*ICML'19*	CLIP(ViT-B-32)	0.17M	10.17	49.73	80.08	18.20	62.47	**95.77**	52.74
CLIP-Adapter [60]	*arXiv'21*	CLIP(ViT-B-32)	0.52M	9.83	42.52	66.79	14.61	51.03	84.14	44.82
Convpass [57]	*arXiv'22*	CLIP(ViT-B-32)	0.14M	16.46	54.79	78.78	14.24	58.67	94.51	52.91
AdaptFormer [58]	*NIPS'22*	CLIP(ViT-B-32)	0.17M	16.92	54.39	77.46	18.74	**63.49**	91.16	53.69
Cross-Modal Adapter [30]	*arXiv'22*	CLIP(ViT-B-32)	0.16M	13.77	50.57	**81.41**	17.43	61.45	95.48	53.62
UniAdapter [31]	*arXiv'23*	CLIP(ViT-B-32)	0.55M	14.46	55.19	**83.95**	16.74	61.43	**95.76**	**54.59**
Ours	-	CLIP(ViT-B-32)	0.16M	**22.71**	**55.81**	80.33	**18.82**	62.84	93.72	**55.71**

features alone on the CLIP, and the methods [57], [58] for vision models can only tune visual features alone. Using the adapter to adapt the representation in a single branch (image or text) of CLIP is suboptimal because it does not allow the flexibility to dynamically adapt both the representation spaces on the RSITR task. UniAdapter and cross-modal adapter are multimodal adapters that support multimodal interaction. However, their structures are complex and fail to consider

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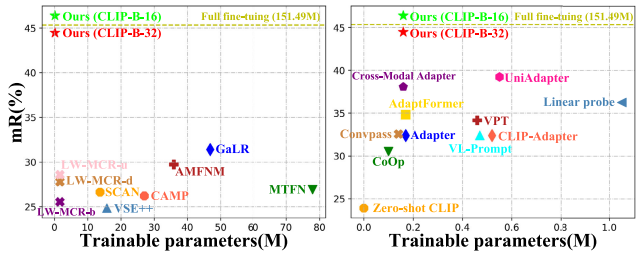


Fig. 8. Retrieval performance versus the number of trainable parameters on the RSITMD dataset. (Left) Comparison to traditional retrieval methods. (Right) Comparison to CLIP-based methods.

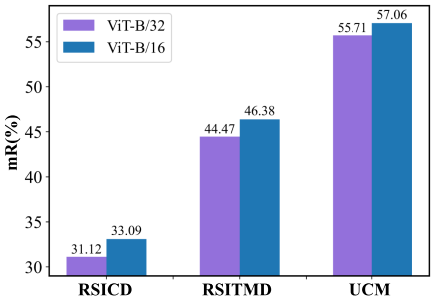


Fig. 9. Investigations with ViT-B/32 and ViT-B/16 backbones of CLIP.

the problem of high intraclass and interclass similarity in RS data. The cross-modal adapter [30] for multimodal tasks converges faster than UniAdapter [31], but the performance of UniAdapter is 1%–2% higher.

The retrieval performance of adapter tuning is 5.9%, 2.4%, and 4.7% higher than that of prompt learning on the three datasets. Prompt learning is an input-related method, and adapter tuning is a network-related method. Due to the significant domain gap between the pretraining CLIP and the RS domain, achieving efficient parameter transfer through adding learnable prompts to the input is limited. To adapt to the different knowledge structures of RS image–text and natural image–text, the adapter can adjust the network structure properly to align the RS image and text modality to obtain significant performance gain.

4) Trained Parameter Efficiency: As shown in Fig. 8 (left), our method requires the smallest trainable parameters compared with the traditional methods. Even though the lightweight model LW-MCR [9] requires 1.65 M parameters, our method requires only 0.16 M to achieve SOTA performance. This suggests that fine-tuning VL pretraining models in the natural domain has great potential for tasks in the RS domain. For the RSITR task, our proposed PETL framework is both parameter-efficient and effective. Among the various PETL methods in Fig. 8 (right), the trainable parameters of our method are not the smallest, but it achieves the optimal performance with 98.9% reduction of the full fine-tuning parameters. Our method (CLIP-B-32) achieves nearly the performance of full fine-tuning on the RSITMD dataset and CLIP-B-16 exceeds it. This shows that it is essential to design the adapter and the PETL framework based on the high intramodal similarity in RS data, which can achieve better performance.

E. Ablation Study

We conduct detailed ablation experiments to validate the effectiveness of the PE-RSITR framework. We systematically analyze the influence of the CLIP backbone network, the proposed MRS-Adapter, and the HMMC loss function on the experimental results. In Sections IV-E1–IV-E6, we conduct experiments mainly on the RSITMD dataset. The performance of T2I retrieval is measured by the sum of $R@K$ ($K = 1$, 5, and 10) of text retrieval, and I2T retrieval is measured by the sum of $R@K$ ($K = 1$, 5, and 10) of image retrieval.

1) Vision Backbones: Fig. 9 shows the results on the three datasets for the two ViT backbones of CLIP, i.e., ViT-B/32 and ViT-B/16. The results are expected: the more advanced the backbone, the better the performance.

TABLE IV
ABLATIONS OF MRS-ADAPTER OF VISUAL AND TEXTUAL BRANCHES

Method	Visual	Textual	T2I retrieval	I2T retrieval	mR
w/o Share		✓	107.58	120.25	37.97
w/o Share	✓		114.51	130.32	40.81
w/o Share	✓	✓	116.47	136.16	42.11
Ours	✓	✓	**128.10**	**138.70**	**44.47**

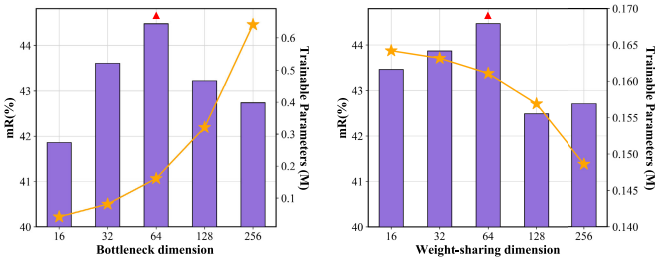


Fig. 10. (Left) Ablations of the bottleneck dimension. (Right) Ablations of the weight-sharing dimension.

2) MRS-Adapter: Since MRS-Adapter involves dual-modal branches and cross-modal interaction, it is necessary to verify that multimodal branches fine-tuned together are more suitable for the task. We applied it to the visual or textual branch alone or to both the branches without the sharing mechanism as comparison experiments. As shown in Table IV, the retrieval performance of applying MRS-Adapter to a single branch is particularly poor, because it cannot adapt the space of the RS domain on both the modalities at the same time. This result proves that MRS-Adapter should be used in both the visual and text branches. When MRS-Adapter is applied to both the branches, the absence of the sharing mechanism can degrade performance. The weight-sharing benefits the realignment of the VL feature space of CLIP in the RSITR task and can reduce the trained parameters. Specifically, MRS-Adapter reduces the parameters of 0.1 M. In summary, MRS-Adapter helps release the power of pretrained CLIP models and improve RSITR performance.

3) Bottleneck Dimension d: The bottleneck dimension d is an important parameter of MRS-Adapter. We conduct experiments with different sizes of bottleneck, as shown in Fig. 10 (left). As d increases, the parameters of MRS-Adapter increase. The retrieval performance is poor when d is less than 64. It is possible that down-projection loses too much information by projecting features to the lower dimensional space. The best performance is achieved when d is 64, and

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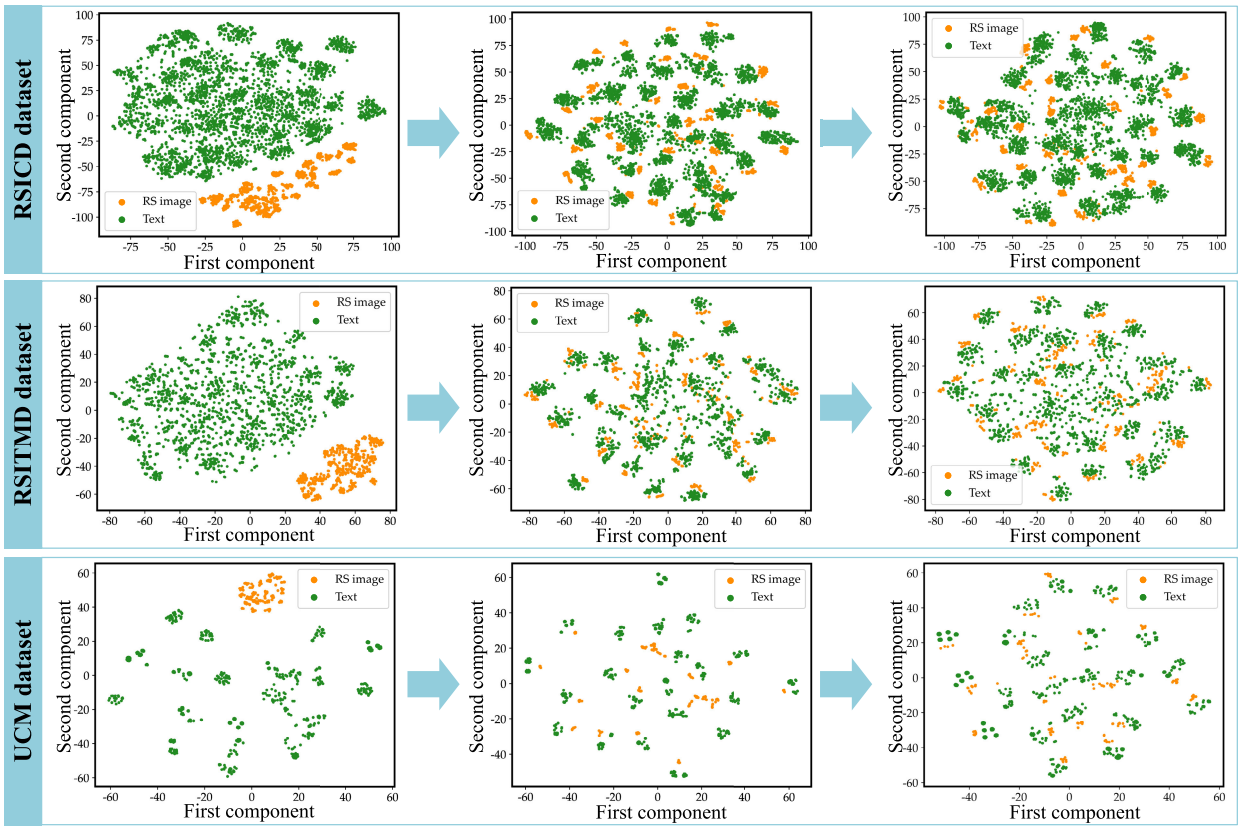


Fig. 11. Detailed change in t-SNE visualizations of RS image and text embeddings before and after transfer learning on three datasets. The first column shows the results of original CLIP before transfer learning, the second shows the results after transfer by MRS-Adapter only, and the third shows the results after adding the framework of intermodal and intramodal cooperative constraints.

the performance starts to decrease as the dimension increases further. Therefore, d of MRS-Adapter is set to 64.

4) Weight-Sharing Dimension r: After the bottleneck dimension d is set to 64, we further explore the impact of the weight-sharing dimension r. We conduct experiments with different sizes of shared weights, as shown in Fig. 10 (right). The overall effect of the weight-sharing dimension on the results is small. The poor retrieval performance when r is small may be due to less knowledge sharing between RS image modality and text modality, which cannot fully facilitate modal alignment. The best performance is achieved when r is 64, and the performance starts to decrease as the dimension increases further. Therefore, r of MRS-Adapter is set to 64.

5) Margins λ and α: In our HMMC learning objective, the margin parameters serve as crucial hyperparameters. To observe the influence of parameters λ, α_v, and α_t on retrieval results, we set up a series of experiments using a controlled variable approach. Regarding the intramodal constraint, we set α_v equal to α_t. As shown in Table V, the results indicate that the model achieves the best retrieval performance, as measured by mR, when the margin parameters λ and α are set to 0.2.

6) HMMC Loss: We have carried out experiments on the HMMC loss, i.e., the intermodal and intramodal cooperative constraint framework. We set up four sets of comparisons to observe the effect of increasing the intramodal constraints on the results, as shown in Table VI. The result shows that the retrieval performance is significantly improved after adding the intramodal constraints. However, when the intramodal constraint is added to the text branch only, text-to-image retrieval becomes worse, while image retrieval has the best result. Likewise, when the intramodal constraint is added to the visual modality only, image-to-text retrieval becomes worse, while text retrieval has the best result. The RSITR task requires both text retrieval and image retrieval. Only adding two intramodal constraints can achieve a balanced result, and the overall retrieval performance is optimal. This result proves that the proposed HMMC loss is effective in further improving the performance on top of the proposed MRS-Adapter.

TABLE V

EXPERIMENT OF THE HMMC LOSS WITH DIFFERENT MARGINS λ AND α

λ	α=α_v=α_t	T2I retrieval	I2T retrieval	mR
0.2	0.1	119.91	137.94	42.97
0.2	0.2	**128.10**	**138.70**	**44.47**
0.2	0.3	121.55	136.57	43.02
0.2	0.4	121.08	134.48	42.51
0.1	0.2	119.68	135.73	42.57
0.2	0.2	**128.10**	**138.70**	**44.47**
0.3	0.2	119.86	136.92	42.80
0.4	0.2	118.72	136.28	42.50

TABLE VI

ABLATIONS OF THE HMMC LOSS

$\mathcal{L}_{intra-v}$	$\mathcal{L}_{intra-t}$	T2I retrieval	I2T retrieval	mR
		113.29	132.14	40.91
	✓	121.76	**141.02**	43.80
✓		**128.47**	134.81	43.88
✓	✓	128.10	138.70	**44.47**

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Fig. 12. Qualitative top-5 text-to-image and image-to-text retrieval results of our PE-RSITR method on the RSITMD dataset. Images with orange bounding boxes are the ground truth and the ground-truth texts are marked with green.

F. Qualitative Results

To get an intuition of how the RS image and text are aligned in the joint embedding space, we present a detailed visualization in Fig. 11. Fig. 11 shows the detailed change in RS image and text embeddings before and after PETL on three datasets. We use the t-distributed stochastic neighbor embedding (t-SNE) method to project the image and text features obtained from the two encoders into a 2-D space. Observing the first column of Fig. 11, the CLIP model of the natural domain before transfer learning exhibits a complete separation of RS image and text modalities due to the domain gap. As shown in the visualization, RSICD has the highest number of sample points. The RSITMD sample points are most scattered and uniformly distributed, indicating the highest fine-grained. The UCM has the lowest number of samples, while different clustering centers indicate different RS scene classes. The unaligned modal representations have semantic inconsistency, and thus, a large number of errors are sure to occur when zero-shot CLIP computes the cross-modal similarity.

To tackle this problem, our MRS-Adapter aggregates the embeddings of each modality into a common space, as shown in the second column of Fig. 11. In addition, it attempts to align image–text pairs separately according to remotely sensed scene-level information to form finer and more discriminative clusters. However, we observe some overlap in each modal representation, which is caused by the great intraclass and interclass similarity. To couple up this problem, we perform the HMMC loss function on top of the proposed MRS-Adapter, as shown in the third column of Fig. 11. The comparison reveals that our method further spars the distribution of multimodal embeddings. The sample points that originally overlapped in a large number of clusters are now clearly visible. In particular, the results of UCM originally had a lot of text clusters without RS image samples next to them, but now there are RS samples near each cluster of text samples. The samples of different modalities are more accurately aligned, alleviating the problem of sample overlap and providing a good representation for cross-modal retrieval and matching.

To qualitatively validate the effectiveness of our PE-RSITR method, we displayed several examples of T2I and I2T retrieval in Fig. 12. Based on these cross-modal retrieval results, our model can accurately distinguish similar samples to retrieve the correct results. Based on these cross-modal retrieval results, our model can accurately distinguish similar samples to retrieve the correct results. The method can understand both the abstract phrases and complex long sentences and is robust in the face of both simple and complex images. This is mainly attributed to the fact that our designed PE-RSITR framework not only learns the specific knowledge of the RS domain but also exploits the powerful generalization ability and rich VL knowledge structure of CLIP.

V. Conclusion and Future Work

In this article, we explore a new paradigm for the PE-RSITR task, namely, PE-RSITR, to bridge the differences between the RS domain and the natural domain. Our proposed MRS-Adapter and the HMMC loss function are conceptually simple but can effectively expand CLIP to coarse-grained VL tasks in the RS domain. Inserting a lightweight MRS-Adapter into CLIP without modifying the inherent structure consumes only a little computational cost. It can make the best use of the powerful generalization ability and rich visual–language prior knowledge of the CLIP. Constructing intramodal positive pairs using random dropout and adding intramodal constraints can effectively avoid the retrieval interference of similar samples. Extensive experiments on RSITR validate that our method can achieve comparable or even better performance than full fine-tuning.

Manual selection of various dimension parameters is required for the MRS-Adapter. How to adaptively select the parameters of the MRS-Adapter remains a challenging topic.

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Our proposed method does not account for scale variations, cluttered backgrounds, and sparse or dense distribution of objects in RS imagery. In future work, we will extend our method to other fine-grained perception tasks, such as RS visual grounding (object level) and RS image change captioning (spatiotemporal level). In these tasks, it is essential to extract fine-grained features or similarities from image–text pairs. We hope our work will inspire future research to explore more efficient PETL methods for more fine-grained RS visual–language tasks.

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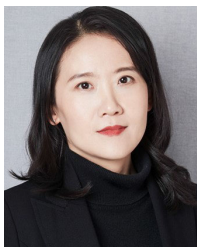
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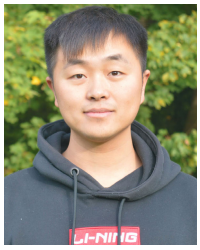


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